PVC Suspension Reactor Ver 1 · R-101 (one of four 140 m³ batch suspension-PVC reactors)

Instrument list and I/O count

PVC-V1-IN-INL-001

Instrument list of R-101 generated from the PVC Suspension Reactor Ver 1 simulator: 66 tags (16 SIS, 50 BPCS), I/O count 59 BPCS + 20 SIS points before a 20 % spare, range, independence and valve-sizing checks, 13 of 20 cross-checks passing and 9 engineering queries.

Unit	PVC Suspension Reactor Ver 1 · R-101 (one of four 140 m³ batch suspension-PVC reactors)
Prepared for	Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University
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Originator	Apinan Soottitantawat

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The numbers in this document come from the PVC Suspension Reactor Ver 1 simulator model (R-101) and the documents of the same OTS, or are labelled "assumed" or "typical". It was generated for teaching: it is not a design deliverable and must not be used for construction or purchase. Checked and Approved are left blank on purpose.

1 Scope and basis

This instrument list covers reactor R-101 of PVC Suspension Reactor Ver 1 (one of four identical 140 m³ batch suspension-PVC reactors): 66 tags in 16 SIS and 50 BPCS lines, with the I/O count, range and valve checks and the engineering queries. It is generated from the simulator: the DCS bubbles and indicators give the measurements, makeCtrls() the loops, ranges and tuning, ALM the alarm limits, tripConditions() and K the trip settings, the fail map inside the model's step() and the P&ID tab table the fail positions, the batch logic the on-off valves, and the text of PVC-PID-001/002 the sheet of each tag.

• Design values: the design state of the OTS, 2.0 h into the reaction (57.0 °C, 8.6 bar g, 26 % conversion).

• Envelope: the reference batch (end point at 5.27 h, X 0.848, 43.2 t PVC), from 15 min after initiation.

• Measuring principles, I/O assignment, card sizes and the 20 % spare are typical selections (assumed). Standards quoted as typical practice: ISA-5.1 (symbols and tags), ISA-18.2 (alarms), IEC 61511 (SIS), IEC 60534-2-1 (valve sizing), NAMUR NE43 (signal levels).

• Related documents: PVC-PID-001/002 (EPC P&ID), PVC-V1-IN-CE-001 (C&E matrix), PVC-V1-IN-SRS-001 (SRS), PVC-V1-SA-HZP-001 (HAZOP report), PFD-100.

2 Deliverables of the I&C discipline

Deliverable	What it holds	On R-101	Where in the OTS
Instrument index / list	One line per tag: service, range, alarms, trips, I/O, fail position, P&ID sheet (this document)	66 tags	this tab
Instrument datasheets	One per device: process data (from the HMB), range, materials, connections, certification (SIL, hazardous area)	40 field devices	Mechanical tab (datasheet form)
Loop diagrams	Wiring of each loop from the device through junction box and marshalling to the I/O card	11 control loops, 42 wired devices	section 4 (the loop explorer draws the functional loop)
I/O list	Every I/O point with its system, card and channel	79 points before spare	section 3
Cause & effect (C&E) matrix	Which cause (trip, alarm, permissive) moves which final element	30 causes × 17 effects	C&E matrix tab (PVC-V1-IN-CE-001)
Logic / sequence diagrams	The batch sequence and the SIS logic drawn as function blocks	7 sequence steps + 4 SIFs	Batch & SIS tab (live), SRS of SIF tab
Alarm list	Every alarm with its limit, priority and rationale (ISA-18.2)	11 DCS alarm tags + C&E events	DCS console, C&E matrix tab
Safety requirements specification (SRS)	Each SIF: sensor, logic, final element, SIL, response time, proof test	4 SIFs	SRS of SIF tab (PVC-V1-IN-SRS-001)
Control narrative and DCS configuration	How each loop and the sequence behave; displays, faceplates, tuning	11 controllers	Control loop design tab, Operate tab
Hook-up, cable schedule, JB layout	How each device is mounted and cabled	—	Electrical tab

2.1 Tag letters (ISA-5.1) used on R-101

Letter	Measured variable	R-101 tags
F	Flow	FI-101, FI-102, FI-103, FI-105, FI-110, FI-120, FI-130, FI-135, FIC-101, FIC-102
H	Hand	HIC-105, HIC-135, HS-001, HV-135
J	Power	JI-100, JT-100
K	Time, schedule	KX-100
L	Level	LI-100, LIC-100, LSHH-100, LT-100

Letter	Measured variable	R-101 tags
P	Pressure	PCV-112, PCV-141, PD-100, PI-100, PI-101, PSHH-101, PSV-100, PT-100, PT-101, PV-105
Q	Quantity	QI-100
T	Temperature	TI-100, TI-101, TI-110, TI-111, TI-120, TI-121, TI-131, TI-132, TIC-100, TIC-110
V	Vibration, machinery analysis	VPC-130
W	Weight, force	WI-100, WT-100
X	Unclassified	XA-133, XI-100, XS-101, XV-101, XV-102, XV-103, XV-104, XV-105, XV-106, XV-108

Non-ISA first letters kept from the DCS: KX-100 (K = time in ISA-5.1) is the K-value estimate, VPC-130 (V = vibration) is a valve-position controller, QI-100 (Q = quantity) is a heat duty. See engineering query on the letters.

3 Instrument list — SIS instruments and final elements

Tag	Service	Type	Range	Unit	Alarms	Trip	I/O	Signal	Fail	P&ID	Notes
LT-100	R-101 level	transmitter	0 – 100	%	L 74 / H 88	LSHH-100 HH 94 %, 5 s	AI + AI (SIS, via signal splitter)	4–20 mA HART	—	001	Non-contact radar on the top head (P&ID: LT-100 RADAR); carries the SIS trip LSHH-100 on the LIC-100 control transmitter (drawing basis)
TT-101	R-101 temperature, independent SIS element TE-101	transmitter	20 – 80	°C	H 60	TSHH-101 HH 63 °C, 10 s	AI (SIS)	4–20 mA HART	Trip on fault (out of range = trip)	001	RTD Pt100 in its own thermowell and nozzle, HART transmitter, SIS-certified · element TE-101
FT-110	Jacket circulation	transmitter	0 – 600	m³/h	L 300	FSLL-110 LL 120 m³/h, 30 s	AI (SIS), repeated to the DCS	4–20 mA HART	—	002	Electromagnetic flowmeter on the jacket loop (assumed); one transmitter for the SIS trip FSLL-110, the DCS alarm and the batch permissive
PT-101	R-101 pressure, SIS transmitter PT-101	transmitter	0 – 16	bar g		PSHH-101 HH 12.5 bar g, 5 s	AI (SIS)	4–20 mA HART	Trip on fault (out of range = trip)	001	Gauge-pressure transmitter on its own nozzle, flush diaphragm seal, SIS-certified
FV-103	Shrinkage-compensation water injection	control valve	0 – 100	% open			AO + DO (SIS solenoid)	4–20 mA HART + 24 V DC solenoid	FC	001	Positioner with HART; fail FC; the SIS forces it closed through a de-energise-to-trip solenoid
TV-110A	Jacket inlet temperature — LP steam	control valve	0 – 100	% open			AO + DO (SIS solenoid)	4–20 mA HART + 24 V DC solenoid	FC	002	Positioner with HART; fail FC; the SIS forces it closed through a de-energise-to-trip solenoid; split range from TIC-110 via TY-110 (0–50 % output)
TV-110B	Jacket inlet temperature — CW injection	control valve	0 – 100	% open			AO + DO (SIS solenoid)	4–20 mA HART + 24 V DC solenoid	FO	002	Positioner with HART; fail FO; the SIS forces it fully open through a de-energise-to-trip solenoid; split range from TIC-110 via TY-110 (50–100 % output)
FV-120	Cooling water to the four baffles	control valve	0 – 100	% open			AO + DO (SIS solenoid)	4–20 mA HART + 24 V DC solenoid	FO	002	Positioner with HART; fail FO; the SIS forces it fully open through a de-energise-to-trip solenoid
FV-130	Cooling water to reflux condenser E-101	control valve	0 – 100	% open			AO + DO (SIS solenoid)	4–20 mA HART + 24 V DC solenoid	FO	002	Positioner with HART; fail FO; the SIS forces it fully open through a de-energise-to-trip solenoid
XV-101	Water charge valve	on-off valve	—	open / closed		SIS: close	DO (BPCS) + DO (SIS) + 2 DI	24 V DC solenoid; limit switches ZSO/ZSC	FC	001	Opened by the batch sequence, closed by the SIS (two solenoids in series)
XV-102	VCM charge valve	on-off valve	—	open / closed		SIS: close	DO (BPCS) + DO (SIS) + 2 DI	24 V DC solenoid; limit switches ZSO/ZSC	FC	001	Opened by the batch sequence, closed by the SIS (two solenoids in series)
XV-104	Short-stop injection valve from pot D-105	on-off valve	—	open / closed		SIS: open	DO + 2 DI (SIS)	24 V DC solenoid; limit switches ZSO/ZSC	FC	001	Opened by the SIS (short-stop): energise to trip, while drawn fail-closed; model: not closed on instrument-air failure
XV-108	Emergency vent to dump tank D-103 (ESD)	on-off valve	—	open / closed		SIS: open (ESD only)	DO + 2 DI (SIS)	24 V DC solenoid; limit switches ZSO/ZSC	FC	001	Opened by the SIS (ESD vent): energise to trip, while drawn fail-closed; model: not closed on instrument-air failure
XV-109	Vacuum valve to VP-101	on-off valve	—	open / closed		SIS: close	DO (BPCS) + DO (SIS) + 2 DI	24 V DC solenoid; limit switches ZSO/ZSC	FC	001	Opened by the batch sequence, closed by the SIS (two solenoids in series)
XS-101	Agitator M-101 running (motor-run contact from the MCC)	switch	—			XS-101 stopped, 15 s	DI (SIS)	Dry contact, 24 V DC	Open contact = trip	001	Sees the motor, not the shaft: a broken coupling is not detected (SRS SIF-2; a shaft-speed switch SS-101 is the SRS proposal)
HS-001	Emergency shutdown push-button	hand switch	—				DI (SIS)	Dry contact, normally closed, 24 V DC	Open = trip	SIS table (001, 002)	Hard-wired to the SIS logic solver; trips as the short-stop SIFs and opens XV-108

4 Instrument list — BPCS instruments, valves and DCS functions

Tag	Service	Type	Range	Unit	Alarms	Trip	I/O	Signal	Fail	P&ID	Notes
FT-102	VCM charge from P-103	transmitter	0 – 250	t/h			AI	4–20 mA HART	—	001	Coriolis mass flowmeter, VCM liquid (t/h, assumed)
FT-101	Demineralised water charge	transmitter	0 – 300	m³/h			AI	4–20 mA HART	—	001	Electromagnetic flowmeter, demineralised water (assumed)
FT-103	Shrinkage-compensation water injection	transmitter	0 – 10	m³/h	H 8		AI	4–20 mA HART	—	001	Electromagnetic flowmeter, small bore (assumed)
JT-100	Agitator M-101 power	transmitter	0 – 450	kW	L 150 / H 420		AI	4–20 mA HART	—	001	Power transducer in the M-101 MCC cubicle (kW)
PT-100	R-101 pressure	transmitter	0 – 16	bar g	H 11		AI	4–20 mA HART	—	001	Gauge-pressure transmitter, flush diaphragm seal (polymer service)
TT-100	R-101 temperature	transmitter	0 – 100	°C	L 56 / H 58		AI	4–20 mA HART	—	001, 002	RTD Pt100 (3-wire) in a flanged thermowell, HART head transmitter · element TE-100
XI-100	Calorimetric conversion	calculated	0 – 100	%			—	DCS calculation	—	001	DCS calculation: Calorimetric conversion: ∫(jacket + baffle + E-101 duty − agitator + stored heat) dt ÷ (ΔHp × FQI-102)
TT-111	Jacket outlet	transmitter	20 – 100	°C			AI	4–20 mA HART	—	002	RTD Pt100 in a thermowell, jacket return · element TE-111
TT-110	Jacket inlet temperature	transmitter	0 – 100	°C	H 75 (hotinit)		AI	4–20 mA HART	—	002	RTD Pt100 in a thermowell after the steam mixer SM-101 · element TE-110
FT-120	Cooling water to the four baffles	transmitter	0 – 250	m³/h	L 15		AI	4–20 mA HART	—	002	Electromagnetic flowmeter, baffle CW supply (assumed)
TT-121	Baffle cooling water return	transmitter	20 – 80	°C			AI	4–20 mA HART	—	002	RTD Pt100 in a thermowell, baffle CW return · element TE-121
FT-130	Cooling water to reflux condenser E-101	transmitter	0 – 600	m³/h			AI	4–20 mA HART	—	002	Electromagnetic flowmeter, E-101 CW supply (assumed)
TT-131	E-101 cooling water return	transmitter	20 – 80	°C	H 58		AI	4–20 mA HART	—	002	RTD Pt100 in a thermowell, E-101 CW return · element TE-131
TT-132	E-101 top	transmitter	20 – 80	°C	L 46		AI	4–20 mA HART	—	001	RTD Pt100 in a thermowell at the E-101 top (inert vent) · element TE-132
FI-135	E-101 inert vent to VCM recovery	calculated	0 – 20	kmol/h			—	DCS calculation	—	001	DCS calculation: Inert-vent flow shown on the faceplate, estimated from the valve position and pressures (no FT drawn on PVC-PID-001)
FI-105	Degassing: unreacted VCM to recovery through PV-105	calculated	0 – 40	t/h			—	DCS calculation	—	001	DCS calculation: Degassing flow shown on the faceplate, estimated from the valve position and pressures (no FT drawn on PVC-PID-001)
TT-120	Cooling water supply temperature	transmitter	20 – 50	°C	H 35		AI	4–20 mA HART	—	002	RTD Pt100 in a thermowell, CW supply header · element TE-120
WT-100	R-101 contents by load cells	transmitter	0 – 160	t			AI	4–20 mA HART	—	001	Three load cells under the brackets + summing weighing transmitter · element WE-100A/B/C
FQI-101	Water charged this batch	calculated	0 – 80	t			—	DCS calculation	—	DCS only	DCS calculation: Totaliser of FT-101 in the DCS; the batch preset closes XV-101
FQI-102	VCM charged this batch	calculated	0 – 80	t			—	DCS calculation	—	DCS only	DCS calculation: Totaliser of FT-102 in the DCS; the batch preset closes XV-102
QI-100	Heat removed	calculated	0 – 5	MW			—	DCS calculation	—	DCS only	DCS calculation: Heat removed = Σ flow × cp × ΔT of jacket, baffles and E-101

Tag	Service	Type	Range	Unit	Alarms	Trip	I/O	Signal	Fail	P&ID	Notes
PD-100	Pressure drop below the running pressure	calculated	0 – 4	bar	H 1.8 (end)		—	DCS calculation	—	DCS only	DCS calculation: Running maximum of PI-100 (30–72 % conversion) minus PI-100; end point at the set drop
KX-100	K-value estimate from the temperature history	calculated	60 – 74				—	DCS calculation	—	DCS only	DCS calculation: K-value estimate from the temperature history of each kilogram of PVC made
FIC-101	Demineralised water charge	controller	0 – 300	m³/h			—	DCS function block	FC	001	PID → FV-101; mode AUTO; Kc 0.3, Ti 4 s, reverse acting
FIC-102	VCM charge from P-103	controller	0 – 250	t/h			—	DCS function block	FC	001	PID → FV-102; mode AUTO; Kc 0.3, Ti 4 s, reverse acting
FIC-103	Shrinkage-compensation water injection	controller	0 – 10	m³/h			—	DCS function block	FC	001	PID → FV-103; mode CAS (set point from LIC-100); Kc 0.3, Ti 4 s, reverse acting
LIC-100	R-101 level	controller	0 – 100	%			—	DCS function block	—	001	PID → FIC-103 set point; mode AUTO; Kc 5, Ti 900 s, reverse acting
TIC-100	R-101 temperature	controller	0 – 100	°C			—	DCS function block	—	001, 002	PID → TIC-110 set point; mode AUTO; Kc 12, Ti 900 s, Td 120 s, reverse acting
TIC-110	Jacket inlet temperature	controller	0 – 100	°C			—	DCS function block	FC / FO	002	PID → TV-110A steam (0–50 %) / TV-110B CW (50–100 %); mode CAS (set point from TIC-100); Kc 2, Ti 40 s, direct acting
FIC-120	Cooling water to the four baffles	controller	0 – 250	m³/h			—	DCS function block	FO	002	PID → FV-120; mode AUTO; Kc 0.3, Ti 4 s, reverse acting
FIC-130	Cooling water to reflux condenser E-101	controller	0 – 600	m³/h			—	DCS function block	FO	002	PID → FV-130; mode CAS (set point from VPC-130); Kc 0.3, Ti 4 s, reverse acting
VPC-130	Valve-position controller: holds the jacket CW valve TV-110B near 50 % by moving the E-101 cooling water ; anti-foam limit 12 m³/h per % of XI-100 conversion	controller	0 – 100	%			—	DCS function block	—	002	PID → FIC-130 set point; mode AUTO; Kc 0.3, Ti 900 s, direct acting
HIC-135	E-101 inert vent to VCM recovery	hand controller	0 – 100	%			—	DCS function block	FC	001	Hand controller → HV-135; mode MAN; operator sets the opening
HIC-105	Degassing: unreacted VCM to recovery through PV-105	hand controller	0 – 100	%			—	DCS function block	FC	001	Hand controller → PV-105; mode MAN; operator sets the opening
FV-101	Demineralised water charge	control valve	0 – 100	% open			AO	4–20 mA HART	FC	001	Positioner with HART; fail FC
FV-102	VCM charge from P-103	control valve	0 – 100	% open			AO	4–20 mA HART	FC	001	Positioner with HART; fail FC
HV-135	E-101 inert vent to VCM recovery	control valve	0 – 100	% open			AO	4–20 mA HART	FC	001	Positioner with HART; fail FC
PV-105	Degassing: unreacted VCM to recovery through PV-105	control valve	0 – 100	% open			AO	4–20 mA HART	FC	001	Model name HV105. Positioner with HART; fail FC
TY-110	Split range: 0–50 % → TV-110A steam, 50–100 % → TV-110B CW	calculated	0 – 100	%			—	DCS function block	—	002	DCS function block between TIC-110 and the two valves (PVC-PID-002)
XV-103	Initiator dosing valve (from pot D-104)	on-off valve	—	open / closed			DO + 2 DI	24 V DC solenoid; limit switches ZSO/ZSC	FC	001	Opened and closed by the batch sequence; model: not closed on instrument-air failure
XV-105	Degassing isolation to VCM recovery (upstream of PV-105)	on-off valve	—	open / closed	event (live)		DO + 2 DI	24 V DC solenoid; limit switches ZSO/ZSC	FC	001	Opened and closed by the batch sequence
XV-106	Bottom discharge valve to blowdown tank D-102	on-off valve	—	open / closed			DO + 2 DI	24 V DC solenoid; limit switches ZSO/ZSC	FC	001	Opened and closed by the batch sequence

Tag	Service	Type	Range	Unit	Alarms	Trip	I/O	Signal	Fail	P&ID	Notes
PCV-141	Nitrogen pad regulator on R-101 (set 1.5 bar g)	regulator	—				—	self-acting	—	001	Self-acting, no I/O; set point on the drawing
PCV-112	Jacket-loop back-pressure regulator, spill to CWR (4 bar g)	regulator	—				—	self-acting	—	002	Self-acting, no I/O; set point on the drawing
M-101	Agitator motor (330 kW)	motor control	—				2 DO + 2 DI	Start / stop pulses; running and tripped contacts (MCC)	Stops on loss of power	001	MCC interface (typical); the running contact of M-101 is repeated to the SIS as XS-101
P-101A	Jacket circulation pump A	motor control	—				2 DO + 2 DI	Start / stop pulses; running and tripped contacts (MCC)	Stops on loss of power	002	MCC interface (typical); the running contact of M-101 is repeated to the SIS as XS-101
P-101B	Jacket circulation pump B (spare)	motor control	—				2 DO + 2 DI	Start / stop pulses; running and tripped contacts (MCC)	Stops on loss of power	002	MCC interface (typical); the running contact of M-101 is repeated to the SIS as XS-101
P-103	VCM charge pump	motor control	—				2 DO + 2 DI	Start / stop pulses; running and tripped contacts (MCC)	Stops on loss of power	001	MCC interface (typical); the running contact of M-101 is repeated to the SIS as XS-101
XA-133	Foam / PVC carried into E-101	calculated	—		event (foam)		—	DCS calculation	—	DCS only	DCS alarm computed from other measurements
PSV-100	Relief valve lifting to dump tank D-103	calculated	—	bar g	H 15.0 (psv)		—	DCS calculation	—	001	Lift alarm computed from PI-100 above the set pressure; no lift switch or rupture-disc indicator is drawn

5 I/O count

Transmitter = 1 AI (4–20 mA HART); control valve = 1 AO, plus 1 SIS DO for a solenoid where the SIS forces the valve; on-off valve = 1 DO per system that drives it + 2 DI limit switches; motor = 2 DO (start, stop) + 2 DI (running, tripped). Totals before spare: BPCS 59, SIS 20 points.

System	Type	Used	Spare (20 %)	Installed	Cards × channels (typical)
BPCS	AI	16	4	20	2 × 16
BPCS	AO	9	2	11	2 × 8
BPCS	DI	20	4	24	1 × 32
BPCS	DO	14	3	17	2 × 16
SIS	AI	4	1	5	1 × 16
SIS	AO	0	0	0	—
SIS	DI	6	2	8	1 × 32
SIS	DO	10	2	12	1 × 16

6 Instrument checks

6.1 Range against design value, batch envelope, alarms and trips

Tag	Range	Unit	Design	% span	Reference batch	Alarm / trip limits	Limits inside	30–70 %	Reads every phase
FT-102	0 – 250	t/h	0.0	0 %	0.0 – 0.0	—	—	—	✓
FT-101	0 – 300	m³/h	0	0 %	0 – 0	—	—	—	✓
FT-103	0 – 10	m³/h	3	32 %	3 – 6	alarm H 8	✓	✓	✓
JT-100	0 – 450	kW	290	64 %	271 – 338	alarm L 150, alarm H 420	✓	✓	✓
PT-100	0 – 16	bar g	8.62	54 %	6.81 – 8.69	alarm H 11	✓	✓	✗ evacuation −0.98
LT-100	0 – 100	%	82.9	83 %	82.5 – 83.9	alarm L 74, alarm H 88, trip HH 94	✓	outside	✓
TT-100	0 – 100	°C	57.0	57 %	56.7 – 57.3	alarm L 56, alarm H 58	✓	✓	✓
TT-101	20 – 80	°C	57.0	62 %	56.7 – 57.3	alarm H 60, trip HH 63	✓	✓	✓
XI-100	0 – 100	%	25.9	26 %	1.1 – 84.6	—	—	—	✓
TT-111	20 – 100	°C	40.9	26 %	40.4 – 49.2	—	—	outside	✓
TT-110	0 – 100	°C	38.2	38 %	37.7 – 47.0	alarm H 75	✓	✓	✓
FT-110	0 – 600	m³/h	473	79 %	473 – 473	alarm L 300, trip LL 120	✓	outside	✓
FT-120	0 – 250	m³/h	80	32 %	80 – 80	alarm L 15	✓	✓	✓
TT-121	20 – 80	°C	36.3	27 %	36.3 – 36.4	—	—	outside	✓
FT-130	0 – 600	m³/h	36	6 %	0 – 91	—	—	outside	✓
TT-131	20 – 80	°C	55.0	58 %	47.4 – 56.8	alarm H 58	✓	✓	✓
TT-132	20 – 80	°C	54.0	57 %	46.0 – 54.3	alarm L 46	✓	✓	✓

Tag	Range	Unit	Design	% span	Reference batch	Alarm / trip limits	Limits inside	30–70 %	Reads every phase
FI-135	0 – 20	kmol/h	0.0	0 %	0.0 – 0.0	—	—	—	✓
FI-105	0 – 40	t/h	0.0	0 %	0.0 – 0.0	—	—	—	✓
TT-120	20 – 50	°C	32.0	40 %	32.0 – 32.0	alarm H 35	✓	✓	✓
PT-101	0 – 16	bar g	8.62	54 %	6.81 – 8.69	trip HH 12.5	✓	✓	✗ evacuation −0.98
WT-100	0 – 160	t	117.8	74 %	111.8 – 133.0	—	—	outside	✓
FQI-101	0 – 80	t	60.0	75 %	60.0 – 60.0	—	—	—	✓
FQI-102	0 – 80	t	51.0	64 %	51.0 – 51.0	—	—	—	✓
QI-100	0 – 5	MW	2.87	57 %	1.73 – 3.54	—	—	—	✓
PD-100	0 – 4	bar	0.00	0 %	0.00 – 1.80	alarm H 1.8	✓	—	✓
KX-100	60 – 74		67	50 %	67 – 67	—	—	—	✓
TY-110	0 – 100	%	—	—	—	—	—	—	✓

• **LT-100** (83 % at design): The radar measures the whole vessel height (0–100 % of the vessel) so that the recipe level 83 %, the alarms and LSHH-100 at 94 % all read on one scale: keep.

• **TT-111** (26 % at design): 0/20–100 °C covers the steam heat-up of the jacket (TIC-110 reaches 75 °C at the start): keep.

• **FT-110** (79 % at design): One pump gives about 473 m³/h, two pumps in parallel about 520 m³/h; 0–600 keeps both on scale: keep.

• **TT-121** (27 % at design): The same range as the other water returns, for interchangeable spares: keep.

• **FT-130** (6 % at design): E-101 takes little water at 2 h; the reference batch reaches about 90 m³/h and the SIS forces FV-130 fully open (≈540 m³/h). A magnetic meter holds ±0.5 % of reading down to about 5 % of span: keep, accept the low normal value.

• **WT-100** (74 % at design): Load cells are sized for the full vessel plus a margin, not for the normal contents: keep.

6.2 Worked example: range to milliamps

I = 4 + 16·(x − LRV)/(URV − LRV). TT-101 (20 – 80 °C) at 57.01 °C gives 13.87 mA; the trip TSHH-101 at 63 °C gives 15.47 mA. PT-101 (0 – 16 bar g) at 8.62 bar g gives 12.62 mA; PSHH-101 at 12.5 bar g gives 16.50 mA. A failed transmitter goes below 3.6 or above 21 mA (NAMUR NE43) and the SIS trips. During evacuation the pressure transmitters would read 3.02 mA (below 4 mA, a fault).

6.3 Independence of the SIS sensors

Trip	Setting	Delay	Armed	Model reads	Drawing: transmitter	Also serves	Own sensor
TSHH-101	HH 63 °C	10 s	always	TI101	TT-101	TI-101	shared with alarm / permissive
PSHH-101	HH 12.5 bar g	5 s	always	PI101	PT-101	PI-101	shared with alarm / permissive
XS-101	motor stopped	15 s	while reacting	M101	XS-101	—	✓
FSLL-110	LL 120 m³/h	30 s	while reacting	FI110	FT-110	FI-110	shared with alarm / permissive
LSHH-100	HH 94 %	5 s	always	LI101	LT-100	LIC-100, LI-100	✗ control transmitter

6.4 Valve sizing from the model coefficients

Liquids: Kv = 3600·C/√(1000·ρ) (m³/h at 1 bar), Cv = 1.156·Kv. Gases, choked (p2 < p1/2): Kv = Qn/(257·p1)·√(ρn·T1), the simplified IEC 60534-2-1 form (typical). Openings assume a linear characteristic.

Valve	Fluid	Model C	ΔP basis	ΔP (bar)	Kv	Cv	Flow at 100 %	Normal flow	Opening
FV-101	Demineralised water 55 °C	45.6 kg/s/√bar	ΔP = water header 6.01 bar a − evacuated R-101 0.31 bar a (start of the charge)	5.70	165.1	190.9	396.7 m³/h	240.0 m³/h	61 %
FV-102	VCM liquid 25 °C	34.0 kg/s/√bar	ΔP = P-103 discharge 13 bar a − R-101 at the start of the VCM charge (water vapour + inerts ≈ 0.47 bar a)	12.53	128.9	149.0	480.3 m³/h	199.6 m³/h	42 %
FV-103	Injection water 30 °C	1.10 kg/s/√bar	ΔP = injection header 16 bar a − R-101 9.63 bar a at design; normal flow = FI-103 at design	6.37	4.0	4.6	10.0 m³/h	3.2 m³/h	32 %
PV-105	VCM vapour to recovery	0.0120 kmol/s/bar	p1 = R-101 9.63 bar a at design, p2 = recovery 1.25 bar a (choked)	8.38	106.6	123.3	24.3 t/h	—	—
HV-135	Inerts (N_2) from the E-101 top	0.0008 kmol/s/bar	p1 = R-101 9.63 bar a, p2 = recovery 1.25 bar a; sized as N_2	8.38	4.8	5.5	25.9 kmol/h	—	—
XV-106	PVC slurry (on-off)	60.0 kg/s/√bar	ΔP at the 2.0 bar g discharge permissive: R-101 3.0 bar a − D-102 1.313 bar a + ≈0.8 bar liquid head	2.50	205.9	238.1	310.5 m³/h	—	—

7 Checks against the model and the other documents

13 of 20 checks pass. Each ✗ is a finding, explained in the detail column and carried as an engineering query in section 8.

#	Check	Against	Result	Detail
1	Every alarm limit inside its transmitter range	ALM (DCS alarm list) · INL ranges	✓	16 alarm limits on 13 instruments, all inside range
2	Every trip setting inside its transmitter range	tripConditions() / K · INL ranges	✓	TSHH-101 63 °C on TT-101 (20–80); PSHH-101 12.5 bar g on PT-101 (0–16); FSLL-110 120 m³/h on FT-110 (0–600); LSHH-100 94 % on LT-100 (0–100)
3	Every SIS trip on its own transmitter, not the control one (drawing basis)	PVC-PID-001/002 · SRS sensors	✗	LSHH-100 on LT-100, the LIC-100 control transmitter (the model trips on its own LI101; SRS proposes its own LT-101). FSLL-110 has FT-110 to itself among the controllers but shares it with the FI-110 alarm and the batch permissive.
4	SIS temperature and pressure sensors independent of the control ones	tripConditions() reads TI101 / PI101 · SRS SIF-1	✓	TT-101 (TI101) vs TT-100 (TI100); PT-101 (PI101) vs PT-100 (PI100)
5	Normal value at design inside the range	snapDesign (2.0 h into the reaction)	✓	27 instruments
6	Normal value at design within 30–70 % of span (typical guideline)	snapDesign · INL ranges	✗	6 of 17 transmitters in service outside: LT-100 83 %, TT-111 26 %, FT-110 79 %, TT-121 27 %, FT-130 6 %, WT-100 74 % — see section 5 for why most of them are kept
7	Range covers the reference batch and the cold-start values	pvBatch reference run · K.Pult (evacuation)	✗	PT-100 reads −0.98 bar g during evacuation, below 0; PT-101 reads −0.98 bar g during evacuation, below 0
8	Every controller in makeCtrls() appears in the INL	makeCtrls()	✓	11 controllers
9	Every bubble on the DCS displays appears in the INL	BUB (DCS console)	✓	21 bubbles
10	Every C&E cause present in the INL (permissives: every instrument they name)	PVC-V1-IN-CE-001 (SF.rows)	✓	30 causes
11	Every C&E effect (final element) present in the INL	PVC-V1-IN-CE-001 (SF.eff)	✓	12 final elements
12	Control-valve fail positions: model FAIL map = P&ID tab loop table	step() FAIL · PD_FAIL	✓	FV-101 FC, FV-102 FC, FV-103 FC, TV-110A FC, TV-110B FO, FV-120 FO, FV-130 FO, HV-135 FC, PV-105 FC
13	Control-valve fail positions: model = PVC-PID-001/002 drawings	step() FAIL · drawings	✓	9 valves agree
14	SIS action = fail position (a de-energise-to-trip solenoid does the job)	bmsTrip() · drawn fail positions	✗	XV-104 opens on trip but is drawn FC; XV-108 opens on trip but is drawn FC (C&E note 12, HAZOP R39). The 5 SIS-forced control valves all go to their fail position.
15	On-off valves: model instrument-air failure = drawn fail-closed	step() air-failure list · drawings	✗	model does not close XV-103, XV-104, XV-108 on air failure
16	Controller tuning on PVC-PID-002 (note 3) = model	PVC-PID-002 note 3 · makeCtrls()	✗	TIC-100 drawn Kc 10, Ti 1500 s, Td 120 s; model Kc 12, Ti 900 s, Td 120 s; TIC-110 drawn Kc 2, Ti 40 s, model Kc 2, Ti 40 s
17	SIS action A on the drawings = the model's trip action	PVC-PID-001 SIS table · bmsTrip()	✗	drawn: close XV-101, XV-102, XV-103; model closes XV-101, XV-102, XV-109 (differ: XV-103, XV-109)
18	Every DCS value has a field device or is flagged as a DCS calculation	BUB / IND · EI_DEVS	✓	11 calculated tags flagged (HIC-135 and HIC-105 show flows no transmitter measures)
19	Every instrument drawn on PVC-PID-001/002 is in the INL	Drawings (text of the SVG sheets)	✓	50 drawn tags (hand valves HV-107, HV-141 left out: manual)
20	Every DCS alarm (ALM) appears on the C&E	ALM · PVC-V1-IN-CE-001	✓	11 alarm tags

8 Engineering queries

No.	Subject	Finding	Raised against	Proposed
EI-01	**LSHH-100 sensor**	PVC-PID-001 draws LSHH-100 on LT-100, the LIC-100 control radar; the model trips on its own level measurement (LI101). The INL lists the drawing, so the SIS trip shares the control transmitter.	PVC-PID-001, PVC-V1-IN-SRS-001 (SIF-4)	Add an independent level transmitter LT-101 for LSHH-100 (SRS proposal LSHHB) and redraw PVC-PID-001.
EI-02	**Pressure transmitter range**	PT-100 and PT-101 are ranged 0–16 bar g but R-101 is evacuated to 0.03 bar a (−0.98 bar g) before every batch; the evacuation permissive reads PI-100.	IND / DCS ranges, PERM-EVAC (C&E)	Range PT-100 as a compound transmitter −1 to 16 bar g (or add an absolute-pressure transmitter for the evacuation); keep PT-101 0–16 bar g for the SIS if the evacuation check stays on PT-100.
EI-03	**TIC-100 tuning**	PVC-PID-002 note 3 gives Kc 10, Ti 1500 s, Td 120 s; the model runs Kc 12, Ti 900 s, Td 120 s (retuned after the drawing was issued).	PVC-PID-002 rev A	Update note 3 to the model tuning at the next revision, or move tuning to the DCS configuration database (tuning is not normally a P&ID note).
EI-04	**Energise-to-trip short-stop and vent valves**	XV-104 and XV-108 must open on a trip but are drawn fail-closed: loss of air or power defeats the action (already in C&E note 12 and HAZOP R39).	PVC-PID-001, PVC-V1-IN-SRS-001	Dedicated air receiver with low-pressure alarm and line monitoring of the solenoid circuits (energise-to-trip design per IEC 61511), or a fail-open design for XV-104 with a separate isolation.
EI-05	**Air failure in the model**	The model does not move XV-103, XV-104, XV-108 on instrument-air failure, while the drawings show them fail-closed.	Simulator model (step())	Model all drawn fail positions on air failure (OTS change request), then rerun the air-failure checks.
EI-06	**SIS action A**	drawn: close XV-101, XV-102, XV-103; model closes XV-101, XV-102, XV-109 (differ: XV-103, XV-109).	PVC-PID-001 SIS table	Correct the SIS table on PVC-PID-001 to the C&E (XV-109 closes, XV-103 is a permissive) at the next revision.
EI-07	**HIC-135 / HIC-105 flows**	The faceplates show the inert-vent flow (kmol/h) and the degassing flow (t/h), but PVC-PID-001 draws no flow element on either line; the values are estimates.	PVC-PID-001, DCS displays	Either add FT-135 / FT-105 (vortex or thermal mass) or label the values as estimated on the displays.
EI-08	**PSV-100 lift alarm**	The C&E lists a PSV-100 lift alarm, but no lift switch or rupture-disc indicator is drawn; the DCS can only infer the lift from PI-100.	PVC-V1-IN-CE-001, PVC-PID-001	Add a lift / burst indicator (e.g. ZSO-100) on PSV-100 or its rupture disc and wire it to the DCS.
EI-09	**Non-ISA first letters**	KX-100 uses K (time, schedule in ISA-5.1) for the K-value, VPC-130 uses V (vibration) for a valve-position controller, and QI-100 (quantity) carries a heat duty.	DCS tag list	Keep the tags (they are on every display and in the training material) but list the meaning in the INL legend; for a new unit use AY-100 (property, computed), ZC-130 and JY-100.